



# FaiST: A Benchmark Dataset for Fairness in Speech Technology

Maliha Jahan<sup>1</sup>, Yinglun Sun<sup>2</sup>, Priyam Mazumdar<sup>2</sup>, Zsuzsanna Fagyal<sup>2</sup>, Thomas Thebaud<sup>1</sup>, Jesus Villalba<sup>1</sup>, Mark Hasegawa-Johnson<sup>2</sup>, Najim Dehak<sup>1</sup>, Laureano Moro-Velazquez<sup>1</sup>

<sup>1</sup>Center for Language and Speech Processing, Johns Hopkins University, Baltimore, MD, USA

<sup>2</sup>University of Illinois Urbana-Champaign, Champaign, IL, USA

mjahan2@jhu.edu, yinglun2@illinois.edu, priyamm2@illinois.edu

## Abstract

Fairness in speech processing systems is a critical challenge, especially regarding performance disparities based on speakers' backgrounds. To help combat this problem, we are introducing FaiST (Fairness in Speech Technology), a novel speech dataset from American English speakers of various racial, ethnic, and national origin groups. The goal is to evaluate and mitigate possible biases across speech technologies using conversational speech from podcasts and interviews online. In FaiST's current version, speakers self-identified as Asian American and African American, and future iterations will include other groups. White American speakers' speech was extracted from VoxCeleb, and their demographic labels were obtained online. In addition to identifiers of race, ethnicity, and national origins, FaiST is also marked for the exact instances in the conversation where self-identifications occurred. We experimented with FaiST and found racial bias in eighteen Automatic Speech Recognition systems.

**Index Terms:** speech, dataset, fairness, bias, speaker demographics, automatic speech recognition

## 1. Introduction

Speech processing systems have become integral to various applications, from virtual assistants to transcription services. However, their performance discrepancy across various speaker demographic groups is a continued topic of interest. Several recent works have demonstrated such performance biases. [1] and [2] have pointed out that some Automatic Speech Recognition (ASR) systems exhibited bias against male speakers. However, studies such as [3] and [4] showed ASR biases against female speakers. Racial and accent biases were also revealed in research works such as [5], [6], and [4]. These performance discrepancies can largely be attributed to biases in training data. To help address this issue, we introduce a novel speech dataset called FaiST (Fairness in Speech Technology). The purpose is to offer speech data from speakers belonging to various racial and ethnic groups to evaluate and mitigate bias.

FaiST dataset, which was collected from the internet, provides human-verified transcripts along with three demographic labels- race, ethnicity, and national origin. These labels are assigned based on speakers either self-identifying or being identified by others, with their implicit or explicit consent, as belonging to those groups. Also, the exact instances in the conversations where these identifications occur are marked in the dataset. In its current version, the dataset mostly contains speakers from Asian American and African American groups, but we are already expanding the dataset to include more groups like Latin American speakers. FaiST also includes a White American group, for which we used the VoxCeleb [7] dataset along with

information from Wikipedia to determine the celebrities' races. We included selected speakers from VoxCeleb into FaiST, and to differentiate the annotation processes, we designate this subset as 'FaiST-V' (FaiST-VoxCeleb), while referring to the rest of the dataset as 'FaiST-EV' (FaiST-Excluding VoxCeleb). The FaiST-EV partition and the metadata necessary to create FaiST-V from VoxCeleb are publicly available.<sup>1</sup>

Several existing datasets have been used to evaluate the biases of speech technologies. Table 1 shows a comparison of multiple such datasets to FaiST. Mozilla's Common Voice [8] has self-reported metadata, but it is not a conversational dataset, nor does it have race and national origin labels. Meta's Casual Conversations [9] is a conversational dataset, but it does not have race and national origin labels. Meta's Fair-Speech [10] has self-reported race labels but does not have national origin labels, and it is not a conversational dataset. In CORAAL, the race labels are not self-reported and it does not feature speakers from a diverse group, mostly focusing on African American speakers. FaiST provides rich, natural conversational speech from a diverse group of speakers, and the FaiST-EV contains self-identified race and national origin labels while also marking the exact identification instances in the conversation.

In this paper, we describe the FaiST dataset and its creation process, from audio collection to annotation. We follow the dataset creation process mentioned in our previous work [11] to create FaiST-EV. Finally, we perform some experiments with eighteen variants of seven ASR systems to uncover any performance biases these ASRs may have. We use the Poisson regression model-based approach of ASR fairness evaluation described in [12]. This method considers speaker variability and confounding effects when evaluating fairness. We use the drop-in-deviance test along with the Poisson regression model to determine the fairness of a model for a particular attribute of a dataset. Our experiments revealed that all eighteen ASRs had racial biases, with the 'white' category performing best twelve out of twenty times.

## 2. Dataset description

The FaiST dataset is an English speech corpus containing natural conversational speech from speakers from various racial and ethnic groups in the U. S. The speakers either self-identify or are identified by others as belonging to those demographic groups. The aim is to enable the scientific community to evaluate the fairness of speech and language technologies or train and develop more equitable and effective speech technologies. Moreover, linguistic researchers can use the dataset to analyze how different groups self-identify through language, as it captures naturalistic speech from diverse populations. FaiST can

<sup>1</sup><https://github.com/Maliha-Jahan/FaiST>

also be used to train Named Entity Recognition (NER) models to find self-identification instances. It is important that speakers self-identify because it ensures that the labels of race, ethnicity, or national origin accurately reflect their personal experience and identity, rather than being imposed or assumed by external sources. It allows for a more authentic and respectful representation of individuals, which is crucial for fairness in datasets and systems that may be influenced by these labels.

### 2.1. Labels

The dataset has race, ethnicity, and national origin labels, as well as spans and types of identification. We followed the U.S. Census Bureau under the U.S. Office of Management and Budget (OMB) to categorize race, ethnicity, and national origin. The categories used in the dataset are described below.

- **Race.** Race attribute has seven categories: White; Black or African American; Asian; American Indian or Alaska Native; Native Hawaiian or Other Pacific Islander; Multiracial; and 'Unknown' for undetermined cases.
- **Ethnicity.** Ethnicity attribute has three classes: Hispanic or Latinx, Non-Hispanic or Non-Latinx, and 'Unknown' for undetermined cases.
- **National Origin Group.** National origin groups were defined as groups of people sharing a common language, culture, ancestry, race, and/or other social characteristics [13]. Some examples are Korean, Vietnamese, Japanese, etc.
- **Type of identification.** "Self-identification" if the speaker identified themselves or "Other-person, in-interaction identification" if someone else present in the interaction identified the speaker.
- **Span.** The exact part of a conversation where a speaker was identified by themselves or others as belonging to a group.

Table 1: Comparison of existing datasets with FaiST. The columns, from left to right, indicate if the content is conversational speech, if the race and national origin labels are available, and if those are self-identified (\* only applies to FaiST-EV)

| Name                        | Conversational | Race | National Origin | Self-Identify |
|-----------------------------|----------------|------|-----------------|---------------|
| CORAAL [14]                 | ✓              | ✓    | x               | x             |
| Common Voice [8]            | x              | x    | x               | N/A           |
| Fair-Speech [10]            | x              | ✓    | x               | ✓             |
| Casual [9]<br>Conversations | ✓              | x    | x               | N/A           |
| FaiST                       | ✓              | ✓    | ✓               | ✓*            |

### 2.2. Statistics

Figure 1 shows the number of speakers (blue bars) and total speech durations (green line) in each race category on the current version of the dataset. The left Y-axis shows the number of speakers. The right Y-axis shows the total hours. There are 526 speakers in total- the largest category being 'White,' and the smallest being 'American Indian or Alaska Native.' The total duration is 145.2 hours. There are 63,409 total segments with the mean and standard deviation of the durations being 8.07 s and 6.43 s. The average number of words per segment is 24.

## 3. Dataset creation

The dataset creation process for most of the White American speakers in FaiST differs from the rest of the dataset. Most of the White speakers were collected from VoxCeleb, whereas, for the rest, we used a semi-automated dataset creation pipeline illustrated in Figure 2. In the following subsections, we will explain both of the processes: FaiST-V and FaiST-EV.

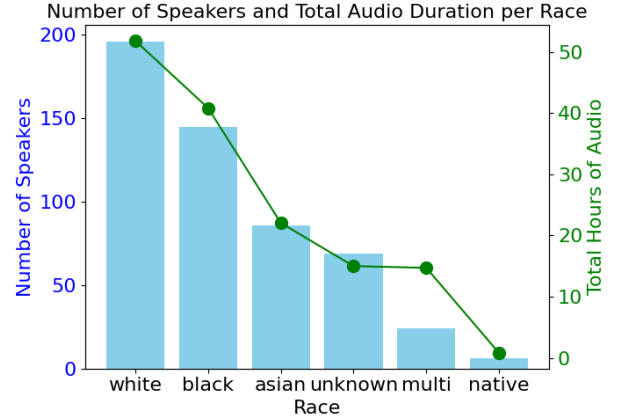


Figure 1: Number of speakers and total duration per race

### 3.1. Dataset collection

For FaiST-EV, we gathered our audio recordings from the Internet Archive<sup>2</sup>. We searched the archive using various keywords focusing on Asian American and African American speakers having discussions and conversations. The keywords are structured following indications included in [11]. We used the names of specific nations to gather data from as many national origin groups as possible. We only downloaded files that are under the CC-BY (Creative Commons Attribution), CC-BY-SA (Creative Commons Attribution-ShareAlike), and CC-BY-NC (Creative Commons Attribution-NonCommercial) licenses to avoid copyright violation issues. For the White American data, we used the VoxCeleb dataset instead of collecting data online, hence creating FaiST-V. We followed this approach because our search for recordings of White American individuals discussing their culture and potentially self-identifying yielded very few results.

### 3.2. Segmentation, diarization, and transcription

After the FaiST-EV audio files were downloaded, they were cleaned to remove segments containing music, environmental noises, etc., automatically, using an audio tagging model [15]. Then, we used a Voice Activity Detection<sup>3</sup> system to keep only speech frames, followed by a speech overlap detection model<sup>4</sup> to remove frames where multiple speakers talk simultaneously. Next, we used a diarization system [16, 17] to identify speaker's turns. Finally, we used a medium-sized Whisper model [18] to transcribe these audio files. For FaiST-V, we directly transcribed the recordings from selected VoxCeleb speakers without the previous steps since these recordings are relatively clean and only have one speaker in them. The speaker selection process is described in section 3.3.2.

<sup>2</sup><https://web.archive.org/>

<sup>3</sup><https://huggingface.co/pyannote/voice-activity-detection>

<sup>4</sup><https://huggingface.co/pyannote/overlapped-speech-detection>

### 3.3. Annotation of race, ethnicity and national origin

#### 3.3.1. Annotation for FaiST-EV

We used GPT-4 to automate the demographic label annotation process following [11]. The annotation tasks performed by GPT-4 can be divided into two parts: Filtering and Tagging.

**Filtering** is the binary task of deciding whether a file is 'relevant' or not. A file is relevant if it is non-fictional, has coherent dialogue or monologue, and talks about race, ethnicity, or national origin. Filtering the files, downloaded as indicated in subsection 3.1, by listening would require many hours for human annotators and would be very costly; hence, we used GPT-4 on the automatic transcriptions of the audio files. The filtering and tagging prompts, obtained from different experiments detailed by [11] are available online<sup>5</sup>.

**Tagging** entails finding instances of identification and labeling the information. GPT-4 is prompted to find any place in the transcription of an audio file where a speaker is being identified by themselves or another speaker in the conversation as belonging to any race/ethnicity/national origin. We call these instances an 'identification'. Whenever GPT finds an identification, it is instructed to gather the following information- source speaker- who is identifying; target speaker- who is being identified; race, ethnicity, national origin- what was the speaker identified as; type- what type of identification it was (as defined in section 2.1), span- the portion of the conversation where this identification happened. The race, ethnicity, or national origin label in one identification only depends on the words spoken in the span of that identification. For instance, if speaker.02 was initially identified as White at the beginning of a conversation but later in the discussion stated, "my Irish parents would..." the second identification would be labeled with the national origin "Irish" but would not include race or ethnicity labels, as only national origin was mentioned in that instance.

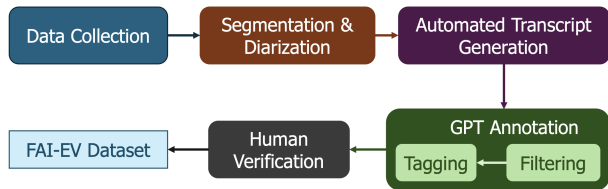


Figure 2: *FaiST-EV dataset creation pipeline*

#### 3.3.2. Annotation for FaiST-V

The demographic annotation and speaker selection process for FaiST-V comes before the transcription step. The process involves reviewing each of the 1,251 celebrities in the development and test sets individually, examining the 'Early Life' or equivalent section in their Wikipedia entries, and recording any available information on their ancestry or descent. Next, we selected speakers who met the following criteria: (1) had available information on their ancestry or descent (which we used as national origin labels), (2) were born in North America, and (3) were of European descent. We ended up with 183 such speakers, whose races we labeled as 'White.' For FaiST-V, we marked ethnicity as 'Unknown,' and we don't have 'Span' labels since there is no self-identification involved.

<sup>5</sup>[https://github.com/Maliha-Jahan/GPT\\_Annotation](https://github.com/Maliha-Jahan/GPT_Annotation)

### 3.4. Human verification

The final step is human verification of the auto-generated transcripts and demographic labels. From all the collected data, we excluded speakers for whom all three labels (race, ethnicity, and national origin) were 'Unknown.' For the rest, for both FaiST-EV and FaiST-V, human annotators listened to the recordings and corrected any mistakes in the transcript. For FaiST-EV, the annotators also verified the automated demographic annotations by removing false positives. False positives are GPT-tagged identifications that are not actually instances of valid identification; for example, GPT may over-interpret someone's racial identity based on names. In the same manner, the annotators ensure that the final demographic label is correct according to the identification found in the transcription and correct it when needed. Our previous work [11] illustrates typical mistakes made by GPT-4 when performing the demographic annotations.

## 4. Experiments

To explore the usability of the FaiST dataset on bias analysis studies, we ran inferences on FaiST using eighteen variants of seven ASR models<sup>6</sup>. On those word error scores, we analyzed racial bias using the Poisson regression model and drop-in-deviance test. The ASR models are described in table 2.

#### 4.1. Evaluation metric

Following the approach in [12], we used a mixed-effect Poisson regression model with rate  $\lambda$ , where word errors serve as the dependent variable, while various attributes (e.g., race) and speaker effects act as independent variables.  $\lambda$  is defined as-

$$\log(\lambda_{ij}) = \log(N_{ij}) + \mu_{f(i)} + \theta^T x_{ij} + r_i \quad (1)$$

In this formulation,  $i$  represents the speaker index, while  $j$  corresponds to the index of the  $i^{th}$  speaker's utterance. The variable  $N_{ij}$  denotes the word count in the utterance. The function  $f(i)$  assigns a group label to speaker  $i$  (e.g., "white" for race), with  $\mu$  capturing the group effect of  $f(i)$ . The term  $x_{ij}$  is a vector representing confounding factors related to the utterance, such as attributes not directly being analyzed for bias, and  $\theta$  quantifies the influence of  $x$  on  $\lambda$ . The random effect at the speaker level,  $r_i$ , follows a normal distribution with a mean of 0 and variance  $\sigma^2$ . It accounts for individual characteristics.

To test the bias for race, we performed the drop-in-deviance ( $DiD$ ) [30] test as shown in Figure 3. By taking the speaker effect and confounding variables as predictors in a simpler model and adding race as an additional predictor in the larger model, we can use  $DiD$  to test whether including the race variable significantly improves the fit of the Poisson model to the data. If it does, then that indicates racial bias is present. This approach accounts for the speaker-level random effects and the effects of any attribute other than race, e.g., national origin. In our experiments, we did not include national origin as a confounding variable because it is an open-ended category with potentially numerous values. We used speaking speed (words per minute) in our experiments as a confounding effect and used speaker labels as a grouping variable to account for the speaker-level random effect. By accounting for these effects, we ensured that the  $DiD$  reflects only race.

$DiD$  is computed as  $-2$  times the difference in the log-likelihood between the simpler and larger models [30]. If the simpler model fits the data perfectly, meaning that adding race

<sup>6</sup><https://github.com/priyammaz/ASRSurvey>

Table 2: Summary of all models used for the ASR experiments. The ‘Mixed Dataset’ for WavLM Large consists of 60K hours of Libri-Light [19], 10K hours of GigaSpeech [20], and 24K hours of VoxPopuli [21]. ‘Emb.’ stands for Embeddings.

| Model                                  | Pretraining Dataset      | ASR Finetuning Dataset | Num. of Parameters |
|----------------------------------------|--------------------------|------------------------|--------------------|
| SEW W2V Small (sew_2) [22]             | Librispeech 960h [23]    | TIMIT [24]             | 89.6 M             |
| SEW W2V Mid (sew_1) [22]               | Librispeech 960h         | Librispeech 100h [23]  | 174.7 M            |
| SEW-D W2V Tiny (swd_3) [22]            | Librispeech 960h         | Librispeech 100h       | 24.1 M             |
| SEW-D W2V Mid (swd_2) [22]             | Librispeech 960h         | Librispeech 100h       | 80.4 M             |
| SEW-D W2V Base (swd_1) [22]            | Librispeech 960h         | Librispeech 100h       | 177.0 M            |
| UniSpeech Large (uni_1) [25]           | CommonVoice [8]          | TIMIT                  | 315.5 M            |
| UniSpeech-SAT Base (uni_2) [26]        | Librispeech 960h         | Librispeech 100h       | 94.3 M             |
| Wav2Vec2 Base (wvc_1) [27]             | Librispeech 960h         | None                   | 94.4 M             |
| Wav2Vec2 Large (wvc_2) [27]            | Librispeech 960h         | None                   | 315.5 M            |
| WavLM Base (wvl_1) [28]                | Librispeech 960h         | Librispeech 100h       | 94.4 M             |
| WavLM Large (wvl_2) [28]               | Mixed Dataset            | Librispeech 100h       | 315.5 M            |
| Whisper Tiny (whs_5) [18]              | 680K h Multilingual [18] | None                   | 37.8 M             |
| Whisper Base (whs_1) [18]              | 680K Hours Multilingual  | None                   | 72.6 M             |
| Whisper Small (whs_4) [18]             | 680K Hours Multilingual  | None                   | 241.7 M            |
| Whisper Medium (whs_3) [18]            | 680K Hours Multilingual  | None                   | 763.9 M            |
| Whisper Large (whs_2) [18]             | 680K Hours Multilingual  | None                   | 1.55 B             |
| Conformer + Relative Emb. (con_3) [29] | Librispeech 960h         | None                   | 618.6 M            |
| Conformer + Rope Emb. (con_4) [29]     | Librispeech 960h         | None                   | 593.3 M            |

in the larger model does not improve the fit, then the  $DiD$  will follow a  $\chi^2$  distribution with degrees of freedom equal to the difference ( $df_{diff}$ ) between the larger and simpler models. The p-value, in this case, is the probability of a  $\chi^2$  random variable with  $df_{diff}$  degrees of freedom being as extreme as the observed  $DiD$  [30]. The p-value suggests whether race is statistically significant in predicting the dependent variables (word error in our case). If it is, then the p-value will be low. So, the higher the p-value, the better the fairness. We used a threshold of 0.01, below which the p-values indicated bias.

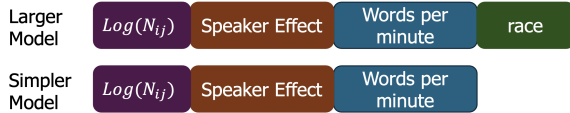


Figure 3: Drop-in-Deviance test.  $N_{ij}$  is the number of words.

## 5. Results and discussion

The p-values for all eighteen experiments were extremely small, which is why they show up as a straight line under the threshold (0.01) in Figure 4. Since the p-values are below the threshold, all eighteen ASRs exhibit racial bias. Figure 4 also shows the coefficients of each race category after fitting a mixed-effect Poisson regression model to the word errors. A higher coefficient means a higher expected word error rate, which signifies a higher bias. From the figure, we can see that the category ‘White’ faces the least bias most of the time (12 out of 18). The other times, the least bias is for the ‘Black’ category. Whisper being trained on multilingual data could be the reason for this.

## 6. Conclusion, limitations, and future work

We introduce FaiST, a novel dataset of conversational speech from a diverse group of speakers. FaiST includes recordings from conversations while containing race, ethnicity, and national origin labels, which were self-identified. FaiST tags the instances in the conversations where the speakers were identified by themselves or someone else as belonging to a group.

These attributes set FaiST apart from the other existing datasets. One of the limitations of the current version of the dataset is that it has different sources of data for FaiST-EV and FaiST-V: though both are interview samples, the recording conditions may differ. Another limitation is that race, ethnicity, and national origin labels are not available for all speakers due to the inherent requirement for the FaiST-EV corpus to contain explicit identifications of speaker demographics. Finally, the speakers in FaiST are interviewers or interviewees in podcasts or TV shows, and these might not represent the breadth of the race or national origin annotated. They may also have higher literacy levels than the broader population they represent. FaiST is an ongoing effort, and we will be releasing newer versions, including more hours of annotated data and other ethnicities.

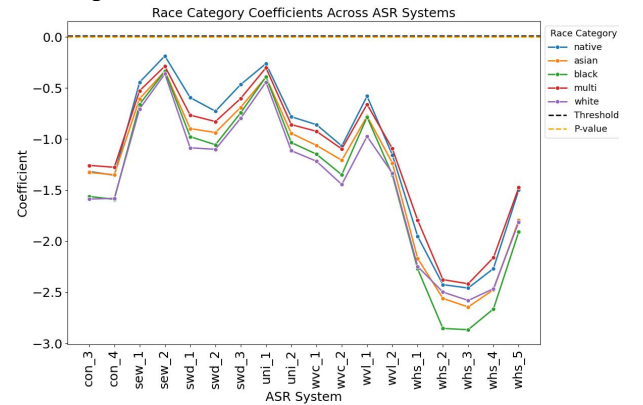


Figure 4: P-values of  $DiD$  and coefficients of race values for all ASRs for FaiST

## 7. Acknowledgements

This project was supported by NSF Award 2147350

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